A QUICK REFERENCE GUIDE TO USING THE NHSN SAS MACROS

Summary

This document describes six SAS macros that have been developed to conduct statistical tests for epidemiological measures, consistent with the methods used by NHSN. Statistical tests are based on the mid-p exact method. Measures tested include common epidemiological measures such as incidence rates, proportions, standardized metrics (standardized infection ratio 'SIR's, standardized utilization ratio 'SUR's, standardized antimicrobial administration ratio 'SAAR's and others) and ratios.

In 2026, NHSN updated functionality of these six macros and developed equivalent Python and R versions comparable to the original SAS implementations.

Overview

Six SAS Macros created by NHSN statisticians are listed below:

Description	Updated Versions (2026) Macro Name	Older Versions Macro Name*
Testing SIR=1	MID-P FOR SIR EQUAL TO 1	SIR (p-value only) SIRcomp (p-value and 95%CI)
Comparing 2 SIRs (ratio)	MID-P FOR COMPARING 2 SIRs	BINOM
95%CI of incidence density rate (single rate)	MID-P FOR INCIDENCE DENSITY RATE	rateCIcomp
Comparing 2 rates (ratio)	MID-P FOR COMPARING 2 RATES	compare_RATES
95%CI of single proportion	MID-P FOR SINGLE PROPORTION	proportionCI
Comparing 2 proportions (ratio)	MID-P FOR COMPARING 2 PROPORTIONS.	Pcomp

*no longer accessible in NHSN

The updated macros are available on the NHSN website (Analysis Resources | NHSN | CDC) and are also accessible via the GitHub repository. Each SAS macro file includes a brief description and

accompanying example data to facilitate user application. In addition, NHSN has developed Python and R implementations that are functionally comparable to the SAS macros, with corresponding documentation and example data provided within each program file. The statistical outputs generated from these implementations have been validated against SAS results.

Brief description, inputs, and corresponding outputs are provided in this document for each of these 6 SAS macro.

1. MID-P FOR SIR EQUAL TO 1

Description
This SAS macro can be used to test whether SIR (or similar standardized metrics like SUR, SAAR) is statistically different from 1, consistent with the methods used by NHSN.

Example dataset and Input parameters
Input parameters:
obs=observed value, exp=predicted value

```
data x;
input obs exp;
cards;
1 2.3
0 3
;
run;

DATA X;SET X;
%sir(obs=obs, exp=exp);
RUN;
proc print;run;
```

Outputs and interpretations

Abbreviations
SIR, p-value, upper limit of 95% CI, lower limit of 95% CI
Note:
*No results will be generated if entry values are missing or impossible values.
*When SIR=0, the lower limit (LLIMIT) of 95%CI is set to missing by default. A user may overwrite LLIMIT=0 if desired.

The SAS System

Obs	obs	exp	midp	SIR	LLIMIT	ULIMIT
1	1	2.3	0.43111	0.43478	0.021754	2.14431
2	0	3.0	0.04979	0.00000	.	0.99858

First example (First row of output):
The p-value calculated by SAS ("midp") is 0.43111. Because this p-value is greater than 0.05, we would conclude that computed SIR value=0.43478 is no statistically different from 1.0. 95%CI also includes the null (1.0)

Second example:
Observed value is 0 so that SIR=0 in which the lower limit is set to missing by default.

REFERENCE:
1. JH ABRAMSON. WINPEPI PROGRAMS. DESCRIBE MANUAL (VERSION 2.42), PAGE-52. Available at 'http://www.brixtonhealth.com/pepi4windows.html'
2. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com.
3. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
4. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.
5. Geoffrey rc, shu-ying y. Mid-p confidence intervals for the poisson expectation. Statistics in medicine, 13,2189-2203 (1994)

2. MID-P FOR COMPARING 2 SIRs

Description
This macro can be used to perform a comparison between two SIRs, consistent with the methods used by NHSN. SIR could be substituted by similar standardized metrics like SUR, SAAR.

Example dataset and input parameters
Data input layout:
o1= observed events in group1
e1= predicted events in group1
o2= observed events in group2
e2= predicted events in group2

```
data exe;
input obs2017 pred2017 obs2016 pred2016;
cards;
10 100 40 50
;
run;
data exe;set exe;
%binom(o1=obs2017, e1=pred2017, o2=obs2016, e2=pred2016);
run;
proc print;run;
```

Outputs and interpretations

Abbreviations
midP= TWOTAILED MID-P value
RATIO= SIR in group2 / SIR in group1 = SIR2/SIR1 = (&O2/&E2)/(&O1/&E1)
LL= 95% lower bound of RATIO
UL= 95% Upper bound of RATIO
Note: When RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

The SAS System

Obs	obs2017	pred2017	obs2016	pred2016	midP	RATIO	LL	UL
1	10	100	40	50	1.8648E-11	8	4.10506	16.8133

SIR for 2017 is 10/100 and SIR for 2016 is 40/50. The ratio of 2 SIRs (2016 SIR/ 2017 SIR) is shown as 8.0 with 95%CI. As the midP value is <0.05, we would conclude that there is significant difference between the two SIRs.

Reference

1. Breslow NE, Day NE. Statistical methods in cancer research. Volume II--The design and analysis of cohort studies. IARC Sci Publ. 1987;(82):1-406. PMID: 3329634. http://www.iarc.fr/en/publications/pdfs-online/stat/sp82/SP82_vol2-3.pdf
2. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959 (Third edition, 2005, Springer Texts in Statistics)
3. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
4. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21

3. MID-P FOR INCIDENCE DENSITY RATE

Description
This macro can be used to perform confidence interval testing around a single incidence density rate (e.g., CLABSI rate), consistent with methods used by NHSN.

Example dataset and input parameters
Input parameters:
numer=numerator, denom=denominator.

```
data z;
input obs persontime;
cards;
0 10
10 10
;
run;
DATA z;SET z;
%rateCIcomp(numer=obs, denom=persontime);
RUN;
proc print;run;
```

Outputs and interpretations

The rate here is multiplied by 1000 so that Incidence density rate is per 1000 person-time & rate_l & rate_u represents.lower and upper limit of 95%CI respectively.
Note: When rate=0, the lower limit (rate_l) of 95%CI is set to missing by default. A user may change it to rate_l=0 if desired.

The SAS System

Obs	obs	persontime	rate	rate_l	rate_u
1	0	10	0	.	299.57
2	10	10	1000	507.940	1782.49

First example (First row of output):
As the incidence density rate=0, the lower limit (rate_l) of 95%CI is set to missing by default.

Second row:
Incidence density rate is 1,000 per 1,000 person-time along with 95%CI.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.

4. MID-P FOR COMPARING 2 RATES

Description

This macro can be used to perform a comparison between two incidence density rates (e.g., CAUTI rates), consistent with the methods used by NHSN.

Example dataset and input parameters

Input parameters

O1= observed events in group-1
PT1=person-time in group-1
O2= observed events in group-2
PT2= person-time in group-2

```
data A;
input Obs1 PTime1 Obs2 PTime2;
cards;
10 200 11 210
;
run;
DATA A;SET A;
%TWORATES(O1=OBS1, PT1=PTIME1, O2=OBS2, PT2=PTIME2);
RUN;
proc print;run;
```

Outputs and interpretations

Abbreviations

mid-P=significance test for comparing between 2 rates
RATE_RATIO = Rate in group-2 / Rate in group-1
LL=lower limit of 95%CI of rate ratio
UL=Upper limit of 95%CI of rate ratio
Note: When RATE_RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

The SAS System

Obs	Obs1	PTime1	Obs2	PTime2	MID_P	RATE_RATIO	LL	UL
1	10	200	11	210	0.91880	1.04762	0.43655	2.53789

As MID_P value is greater than 0.05, we would conclude that there is no significant difference between the two incidence rates The rate ratio between rate in group-2 (Obs2/PTime2) & rate in group-1 (Obs1/PTime1) is 1.04762 with 95%CI which includes the null (1.0).

Reference

1. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959(Third edition, 2005, Springer Texts in Statistics)
2. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
3. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21
4. Austin, Harland (Emory). Epidemiology method-II: Statistical issues for Density type follow-up studies.

5. MID-P FOR SINGLE PROPORTION

Description

This macro can be used to perform confidence interval testing around a single proportion (e.g., healthcare worker flu vaccination percentages). This macro can also be used to compare a single proportion to a benchmark value, consistent with methods used by NHSN

Example dataset and input parameters

Input parameters:
vx=numerator, vn=denominator.

```
data test;
input numerator denominator;
cards;
5 10
;
run;
data test;set test;
%binom(vx=numerator, vn=denominator);
run;
proc print; run;
```

Outputs and interpretations

Abbreviations
vp=proportion, dl=lower limit of 95%CI, du=upper limit of 95%CI
Note: When vp=0, the lower limit (dl) of 95%CI is set to missing by default. A user may change it to dl=0 if desired.

The SAS System

Obs	numerator	denominator	vp	dl	du
1	5	10	0.5	0.21201	0.78799

Point estimate of single proportion of 5/10 is 0.5 with 95%CI provided.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981.
4. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. Statistics in Medicine 1988;17:857-872.
5. Stein Vollset. Confidence intervals for a binomial proportion statistics in medicine,12,809-824 (1993)

6. MID-P FOR COMPARING 2 PROPORTIONS

Description

This macro can be used to perform a comparison of two proportions (2 x 2 contingency table), ie, to compare the risks (prevalence) of outcome in two independent groups.

Example dataset and input parameters

Input parameters:

- exposed with disease (A),
- not-exposed with disease (B),
- exposed without disease (C),
- not-exposed without disease (D).

Letters refer to positions in classic 2X2 exposure/disease status table.

```
Data input layout:
A= E+,D+
B= E-,D+
C= E+,D-
D= E-,D-
                                Disease/Outcome
                          |    Yes                 No            |
                -----------|-------------------------------------|-------
        Exposure     Yes  |     A                   C            |
                     No   |     B                   D            |
                -----------|-------------------------------------|-------
```

```
data x;
input AA BB CC DD;
cards;
-1 20 11 21
10 0 7 20
1 . 3 4
1 2 3 4
;
run;

DATA X;SET X;
%TWOBYTWO(A=AA,B=BB,C=CC,D=DD);
RUN;
PROC PRINT;run;
```

Outputs and interpretations

The SAS System

Obs	AA	BB	CC	DD	MID_P
1	-1	20	11	21	.
2	10	0	7	20	0.00006
3	1	.	3	4	.
4	1	2	3	4	0.83333

Output for second observation (obs=2):
Mid-p=0.00006 indicates that proportion (prevalence, risk) of disease in exposed group is statistically different from that of non-exposed group.

Last example of output (obs=4):
Mid-p=0.83333 indicates that proportion (prevalence, risk) of disease in exposed group is not statistically different from that of non-exposed group.

Other examples have missing cell value or incorrect value such as negative value, mid-P is not computed.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981